

Admitted for suspected deep vein thrombosis of leg, which after investigation is ruled out and no follow-up necessary. Main Condition: Ruled out deep vein thrombosis. Code main condition cluster:

[QA02.Y/BD71.4](#)

Explanation: [QA02.Y](#) indicates that an other specified condition was ruled out and postcoordination allows specification of what the suspected condition was (deep vein thrombosis leg).

Example 2

A child is found playing with an empty acetaminophen bottle. The mother is uncertain if there were any tablets in the bottle. The child is brought to the hospital and following investigation, it is determined that the child did not ingest any pills. Main condition: Ruled out unintentional ingestion of acetaminophen. Code: [QA02.5](#) *Observation for suspected toxic effect from ingested substance, ruled out*. Explanation: In this example, [QA02.5](#) specifies the condition that was ruled out.

2.23.18 Coding of conditions documented as sequela (late effect)

‘Sequelae’ include residual effects of diseases or disorders, injuries or poisonings specified as such, or as late effect of, arrested, cured, healed, inactive, old or quiescent condition unless there is evidence of active disease. Conditions documented as a sequela (late effect) will typically be classified using postcoordination depending on the case.

The cluster should contain:

- first, a stem code identifying the specific manifestation (i.e. nature of the effect), and
- second, a stem code designating ‘late effect of’ (either a code from the body system chapters or a code from Chapter 24 Factors influencing health status or contact with health services)
- third, if required, a stem code representing the prior condition causing the sequelae

Note: Coding of sequelae of an injury with all detail will require four codes in the cluster, the fourth code being the external cause code.

Example 1:

Joint contracture present as a late effect of a prior burn. Code cluster: [FA34.3](#) *Contracture of joint* [QC50](#) *Late effect of prior health problem, not elsewhere classified* [NE11](#) *Burn of unspecified body region* / [PB1Z](#) *Unintentional exposure to unspecified thermal mechanism*

Example 2:

Hemiplegia present as a late effect of old cerebral ischemic stroke. Code cluster: [MB53.Z](#) *Hemiplegia, unspecified* / [8B25.0](#) *Late effects of cerebral ischemic stroke* Note: In Example 2, the concept of late effect and underlying cause is already precoordinated in the stem code [8B25.0](#).

2.23.19 Standards and coding instructions for injury events

The WHO definition of an ‘injury’ is: ‘acute exposure to physical agents such as mechanical energy, heat, electricity, chemicals, and ionizing radiation interacting with the body in amounts or at rates that exceed the threshold of human tolerance. In some cases, (for example, drowning and frostbite), injuries result from the sudden lack of essential agents such as oxygen or heat’. Injuries may be categorised in a number of ways. However, for most analytical purposes and for identifying intervention opportunities, it is especially useful

to categorise injuries according to whether or not they were deliberately inflicted and by whom. Commonly used categories are:

- unintentional (i.e. accidental)
- intentional (i.e. deliberate)
- interpersonal (e.g. assault and homicide)
- self-harm (e.g. abuse of drugs and alcohol, self-mutilation, suicide)
- legal intervention (e.g. action by police or other law enforcement personnel)
- war, civil insurrection and disturbances (e.g. demonstrations and riots)
- undetermined intent

Regarding the collection of events that cause injuries, a set of definitions apply. See section 'Definition related to transport injury events' below.

2.23.19.1 Descriptions related to transport injury events

- (a) A 'transport injury event' is any unintentional injury involving a device designed primarily for, or being used at the time primarily for, conveying persons or goods from one place to another.
- (b) A public highway (trafficway) or street is the entire width between property lines (or other boundary lines). It includes the space of open public land used for purposes of moving persons or property from one place to another. A roadway is that part of the public highway designed, improved and customarily used for vehicular traffic.
- (c) A road traffic injury event is any unintentional injury occurring on the public highway [i.e. originating on, terminating on, or involving a vehicle partially on the highway]. An unintentional injury involving a vehicle is assumed to have occurred on the public highway unless another place is specified, except in the case of unintentional injury involving only off-road motor vehicles, which are classified as unintentional injury not caused by traffic unless the contrary is stated.
- (d) An off-road, nontraffic road injury event is any unintentional injury that occurs entirely in any place other than a public highway.
- (e) A pedestrian is any unintentionally injured person involved who was not at the time of the event riding in or on a motor vehicle, railway train, streetcar or bus or animal-drawn or other vehicle, or on a pedal cycle or animal.
- Pedestrians include those:
 - changing a tire of vehicle
 - making an adjustment to the motor of a vehicle
 - on or in items which assist with pedestrian conveyance, including:

- baby carriage
 - ice-skates
 - perambulator
 - push-chair
 - roller-skates
 - scooter
 - skateboard
 - skis
 - sled
 - wheelchair (powered)
- (f) A driver is an occupant of a transport vehicle who is operating or intending to operate it.
- (g) A passenger is any occupant of a transport vehicle other than the driver.
- Excludes: person traveling on outside of vehicle - see definition (h) below
- (h) A person 'traveling on' a transport vehicle includes any person being transported by a vehicle but not occupying the space normally reserved for the driver or passengers, or the space intended for the transport of property.
- 'Traveling on' includes:
 - bodywork
 - bumper [fender]
 - hanging on outside
 - roof (rack)
 - running-board
 - step
- (i) A pedal cycle is any land transport vehicle operated solely by pedals.
- Includes: bicycle tricycle Excludes: motorised bicycle - see definition (k)
- (j) A pedal cyclist is any person riding on a pedal cycle or in a sidecar or trailer attached to such a vehicle.
- (k) A motorcycle is a two-wheeled motor vehicle with one or two riding saddles and sometimes with a third wheel for the support of a sidecar. The sidecar is considered part of the motorcycle.
- Includes:
 - moped motor scooter motorcycle:
 - NOS
 - combination
 - with sidecar
 - motorised bicycle
 - speed-limited motor-driven cycle
 - Excludes: motor-driven tricycle - see definition (m)

- (l) A motorcycle rider is any person riding on a motorcycle or in a sidecar or trailer attached to such a vehicle.
- (m) A three-wheeled motor vehicle is a motorised tricycle designed primarily for on-road use.
 - Includes:
 - motor-driven tricycle
 - motorised rickshaw
 - three-wheeled motor car
 - Excludes:
 - motorcycle with sidecar - see definition (k)
 - special all-terrain vehicle - see definition (x)
- (n) A car (automobile) is a light transport vehicle with four or more wheels designed primarily for carrying up to 10 persons. A trailer or caravan being towed by a car is considered a part of the car.

Includes: minibus

- (o) A motor vehicle or vehicle may refer to various transport vehicles. The local usage of the terms should be established to determine the appropriate code. If the terms are used ambiguously, use the code for 'unspecified'. A trailer or caravan being towed by a vehicle is considered a part of the vehicle.
- (p) A light goods vehicle (pick-up truck or van) is a motor vehicle with four or more wheels designed primarily for carrying property on roads, weighing less than the local limit for classification as a heavy goods vehicle (usually less than 3500 kg), and not requiring a special driver's licence. A trailer or caravan being towed by a light goods vehicle is considered a part of the vehicle.
- (q) A heavy goods vehicle is a motor vehicle designed primarily for carrying property on roads, meeting local criteria for classification as a heavy goods vehicle in terms of curbside weight (usually above 3500 kg), and requiring a special driver's licence.
- (r) A bus is a motor vehicle designed or adapted primarily for carrying more than 10 persons and requiring a special driver's licence to operate.
- (s) A railway train or railway vehicle is any device, with or without cars coupled to it, designed for traffic on a railway.
 - Includes:
 - interurban:
 - electric car
 - street car (operated chiefly on its own right-of-way, not open to other traffic) railway train, any power [diesel] [electric] [steam]
 - funicular
 - monorail or two-rail subterranean or elevated other vehicle designed to run on a railway track
 - Excludes:

- interurban electric cars (streetcars) specified to be operating on a right-of-way that forms part of the public street or highway - see definition (t)
- (t) A streetcar is a vehicle designed and used primarily for transporting persons within a municipality, running on rails, usually subject to normal traffic control signals, and operated principally on a right-of-way that forms part of the roadway. A trailer being towed by a streetcar is considered a part of the streetcar.
 - Includes:
 - interurban electric car or streetcar, when specified to be operating on a street or public highway
 - tram (car)
 - trolley (car)
- (u) A special vehicle mainly used on industrial premises is a motor vehicle designed primarily for use within the buildings and premises of industrial or commercial establishments.
 - Includes:
 - battery-powered:
 - airport passenger vehicle
 - truck (baggage)(mail)
 - coal-car in mine
 - forklift (truck)
 - logging car
 - self-propelled truck, industrial
 - station baggage truck (powered)
 - tram, truck or tub (powered) in mine or quarry
- (v) A special vehicle mainly used in agriculture is a motor vehicle designed specifically for use in farming and agriculture (horticulture), for example to work the land, tend and harvest crops and transport materials on the farm.
 - Includes:
 - combine harvester
 - self-propelled farm machinery
 - tractor (and trailer)
- (w) A special construction vehicle is a motor vehicle designed specifically for use in the construction (and demolition) of roads, buildings and other structures.
 - Includes:

- bulldozer
 - digger
 - dumper truck
 - earth-leveller
 - mechanical shovel
 - road-roller
- (x) A special all-terrain vehicle is a motor vehicle of special design to enable it to negotiate rough or soft terrain or snow. Examples of special design are high construction, special wheels and tyres, tracks, and support on a cushion of air.
- Includes: - hovercraft on land or swamp - snowmobile
 - Excludes: hovercraft on open water - see definition (y)
- (y) A watercraft is any device for transporting passengers or goods on water.

Includes: hovercraft NOS

- (z) An aircraft is any device for transporting passengers or goods in the air.

2.23.19.2 Classification and coding instructions for unintentional injury caused by transport

Transport injury events are counted for official statistics where they are unintentional.

1. If an event is unspecified as to whether it was a traffic or a nontraffic-injury event, the following definitions should be used:
 - a) Classify as a traffic injury event occurs when the event is classifiable to the traffic categories.
 - b) Classify as a nontraffic injury event occurs when the event is classifiable to nontraffic categories.

For these categories the victim is either a pedestrian, or an occupant of a vehicle designed primarily for off-road use.

2. When unintentional injury involving more than one kind of transport are reported, the following order of precedence should be used:
 - aircraft and spacecraft
 - watercraft
 - other modes of transport
3. Where transport injury event descriptions do not specify the victim as being a vehicle occupant and the victim is described as crushed, dragged, hit, injured, killed, knocked down, run over by any vehicle including:

- animal being ridden
- animal-drawn vehicle
- bicycle
- bulldozer
- bus
- car
- motorcycle
- motorised tricycle
- pick-up (truck)
- recreational vehicle
- streetcar
- tractor
- train
- tram
- truck
- van

Classify the victim as a pedestrian.

4. Where transport injury event descriptions do not indicate the victim's role, classify the victim as an occupant or rider of the vehicle if there is mention of vehicles such as:

- airplane
- bicycle
- boat
- bulldozer
- bus
- car
- motorcycle
- motorised tricycle
- pick-up (truck)
- recreational vehicle
- spacecraft
- tractor
- train
- tram
- truck
- van
- watercraft
- accident
- collision
- crash
- wreck
- NOS

Classify the victim as an occupant or rider of the vehicle mentioned. If more than one vehicle is mentioned, do not make any assumption as to which vehicle was occupied by the victim unless the vehicles are the same. Instead, code to the appropriate categories, taking into account the order of precedence given in note 2 above.

5. Where a transport injury event, such as:

- vehicle (motor) (non-motor): – going out of control (due to):
- burst tyre (blowout)
- driver falling asleep
- driver inattention
- excessive speed
- failure of mechanical part resulted in a subsequent collision

Classify the unintentional injury as a collision. If an unintentional injury other than a collision resulted, classify it as a noncollision injury according to the vehicle type involved.

6. Where an unintentional injury involving a vehicle in motion, such as:
- unintentional poisoning from exhaust gas generated by breakage of any part or explosion of any part of
 - fall, jump or being unintentionally pushed from
 - fire starting in
 - hit by object thrown into or onto
 - injured by being thrown against some part of, or object in
 - injury from moving part of
 - object falling in or on vehicle in motion
 - resulted in a subsequent collision

Classify as a collision.

If an accident other than a collision resulted, classify it as a noncollision injury according to the vehicle type involved.

Unintentional injury due to land transport described as:

- collision (due to loss of control) (on highway) between vehicle and:
 - abutment (bridge)(overpass)
 - fallen stone
 - guard rail or boundary fence
 - inter-highway divider
 - landslide (not moving)
 - object thrown in front of motor vehicle
 - safety island
 - tree
 - traffic sign or marker (temporary)
 - utility pole
 - wall of cut made for road
 - other object, fixed, movable or moving
- overturning (without collision)
- collision with animal (herded)(unattended)
- collision with animal-drawn vehicle or animal being ridden are included.

2.23.20 Conceptual model for quality and patient safety

Exposure to health care events sometimes has unintended and undesired consequences. Health care, the people to whom it is provided, and the complications that can arise in the course of care are highly diverse and complex. Representing them comprehensively in an information system is challenging and is presently beyond the bounds of practicality for routine administrative information systems of the types that are intended to make use of the ICD. The conceptual model has three components:

1. **Harm** to the patient: What was the main consequence for the patient's health?
2. **Cause** or source of harm: What caused the harm?
3. **Mode** or mechanism: In what way? How did the source of harm actually produce harm?

2.23.20.1 Overview of code-set in ICD–11 for quality and patient safety

A key feature of the quality and patient safety code-set in ICD–11 is that a cluster of codes is required to represent a case. Use of the term 'cluster' is novel in ICD–11 and so is the extent and formalisation of the requirement for postcoordination. The quality and safety use case of the ICD is based on the availability of large numbers of methodological tools that are originally based on ICD–10. Specific examples include the Charlson and Elixhauser co-morbidity indices, AHRQ (Agency for Healthcare Research and Quality) Patient Safety Indicators, the Hospital Standardised Mortality Ratio, and various other administrative data quality indicators. WHO recommendations on coding rules for hospital separation episodes are designed to improve comparability of records across hospitals and jurisdictions. Specific examples of coding rules include:

- rules for specifying the main condition,
- numbers of codes per record,
- code clustering mechanisms, and
- use of a status display system that distinguishes diagnoses arising during a hospital stay from those present at admission.

Quality and patient safety reporting is often focused not only on diagnostic information available in the International Classification of Diseases, but also on procedure information, that is currently coded in various country-specific procedure coding systems. The harmonisation of ontological concepts in international procedure coding systems will be important going forward. The available medical and surgical complication codes of ICD–11 are in line with current knowledge in the domain of safety and adverse events.

There are three parts to the Quality and Safety model. The first component, quality and patient safety **Harm**, is usually represented by a standard ICD–11 diagnosis code, from (almost) any chapter of the classification. Some forms of harm that can result from quality and safety events are not adequately represented by a standard ICD–11 diagnosis code. A special set of categories to represent these forms of harm are provided in the injury chapter of ICD–11 (Chapter 22 'Injury, poisoning or certain other consequences of external causes'), under the category titled 'Injury or harm arising from surgical or medical care, not elsewhere classified'. Quality and patient safety causes (sources of harm) fall into four types of causes at the top level that capture events caused by:

1. substances (drugs and medicaments, etc.),
2. procedures,
3. devices, and
4. a mix of other types of causes.

The full quality and safety external cause codes are found in Chapter 23 'External causes of morbidity or mortality' within category titled 'Causes of health care related harm or injury'.

Quality and safety **Mode** or **Mechanism** ('Mode' is the term used in ICD-11 external cause codes) is the second part of the model, and refers to the main way in which the Quality and safety Cause leads to the **Harm** which is represented in the third concept, Quality and safety Harm. Quality and safety **Modes** are specific to the types of Quality and safety **Cause**. Examples are:

Table 1: *Examples of corresponding quality and safety mode or mechanism*

Cause or Source of Harm	Mode or Mechanism
Substance	Overdose, underdose, incorrect substance, or harm arising despite correct administration and dosing
Procedure	Unintentional perforation of an organ during a procedure
Device	Dislodgement, malfunction
Other cause	Mismatched blood; Patient dropped during transfer from operating room table

Examples for the ICD–11 Quality and safety coding model

The ICD–11 quality and safety coding model is demonstrated by the examples in the following table.

Table 2: *Demonstration of the quality and safety model using examples*

Example	Criterion	Detail
1	Case	A woman has been admitted to hospital for stabilisation of diabetes. She is erroneously prescribed three times the usual dose of an antidiabetic medication. The abnormally high dose is given, and the patient has a hypoglycaemic episode
	Harm	Hypoglycaemia in the context of diabetes, unspecified (5A21.Z)
	Cause	Drugs, medicaments or biologic substances associated with injury or harm in therapeutic use (PL00); Medication (use additional code, if desired) - Antidiabetic (XM8S35)
	Mode Cluster	Overdose of substance as mode of injury or harm (PL13.0) 5A21.Z/PL00&XM8S35/PL13.0
2	Case	Patient presented to hospital with visual hallucinations due to malaria prophylaxis with mefloquine prescribed and taken at the correct dose.
	Harm	Visual hallucinations (MB27.27)
	Cause	Drugs, medicaments or biological substances associated with injury or harm in therapeutic use (PL00); Medication (use additional code, if desired) - Mefloquine (XM50J2)
	Mode Cluster	Drug-related injury or harm in context of correct administration or dosage, as mode of injury or harm (PL13.2) MB27.27/PL00&XM50J2/PL13.2
3	Case	A man visits a primary care physician for removal of a skin lump, mainly to exclude the possibility of malignancy. The lesion is excised and the wound is sutured. It later becomes known that the physician had Hepatitis C and the patient has now contracted this disease.
	Harm	Acute hepatitis C (1E50.2)
	Cause	Biopsy procedure, not elsewhere classified, associated with injury or harm in therapeutic use (PK81.5)
	Mode Cluster	Failure of sterile precautions as mode of injury or harm (PL11.4) 1E50.2/PK81.5/PL11.4
4	Case	An elderly woman is admitted due to a fractured neck of femur. Surgical fixation is undertaken. The operative site bleeds heavily the day after surgery, requiring return to theatre.
	Harm	Haemorrhage or haematoma of other or unspecified site complicating a procedure, not elsewhere classified (NE81.0Z) & XA1673 Femoral neck
	Cause	Musculoskeletal procedure associated with injury or harm, open approach (PK80.80) (Orthopaedic surgical procedures are included here)

Example	Criterion	Detail
5	Mode	Unspecified mode of injury or harm associated with a surgical or other medical procedure (PL11.Z) (Note: Select PL11.Z because case documentation does not mention any specific mode or mechanism by which haemorrhage occurred)
	Cluster	NE81.0Z & XA1673 / PK80.80 / PL11.Z
	Case	A 63 year old man had a left knee-replacement less than a year ago, because of arthritis. The implanted device has come loose, resulting in pain and reduced function.
	Harm	Pain in joint (ME82); Specific Anatomy (use additional code, if desired) Knee joint (XA8RL1); Laterality (use additional code, if desired) – Left (XK8G)
	Cause	Orthopaedic devices associated with adverse incidents, prosthetic or other implants, materials or accessory devices (PK99.2)
6	Mode	Dislodgement, misconnection or de-attachment, as mode of injury or harm (PL12.4)
	Cluster	ME82 & XA8RL1 & XK8G / PK99.2 / PL12.4
	Case	A man has bowel cancer. Abdominal surgery was done several days ago to resect the affected part of the colon and re-join the preserved part of the colon. The anastomosis has leaked and required surgical revision.
	Harm	Postsurgical leak (NE81.3) (anastomosis leak is an index term)
	Cause	Gastrointestinal, abdominal, or abdominal wall procedure associated with injury or harm in therapeutic use PK80.3Z
7	Mode	Unspecified mode of injury or harm associated with surgical or other medical procedure (PL11.Z). (Note: Select PL11.Z because case documentation does not mention any specific mode or mechanism by which the anastomotic leak occurred).
	Cluster	NE81.3 / PK80.3Z /[PL11.Z]
	Case	Refractory urinary tract infection due to chronic indwelling catheter.
	Harm	Urinary tract infection, site and agent not specified (GC08.Z)
	Cause	Gastroenterology or urology devices associated with adverse incidents, urinary catheter (PK93.10)
8	Mode	Other specified mode of injury or harm associated with a surgical or other medical device, implant or graft (PL12.Y) (Note: Select PL12.Y because none of the more specific mode types appears to lead to infection of device)
	Cluster	GC08.Z / PK93.10 / PL12.Y
	Case	Elderly patient falls out of bed in a hospital and suffers a left hip fracture. The documentation describes that the nurse forgot to put the bedrails in place which lead to the patients fall.

Example	Criterion	Detail
	Harm	Fracture of neck of femur, unspecified (NC72.2Z ; Laterality (use additional code, if desired) - Left (XK8G)
	Cause	Other health care related causes of injury or harm (PL10)
	Mode	Fall in health care (PL14.E)
	Cluster	NC72.2Z & XK8G/PL10/PL14.E
9	Case	Patient received an infusion of red blood cells and develops severe rigors that subside after an hour. It was discovered that there was a blood mismatch (not ABO or Rh incompatibility).
	Harm	Other serum reactions (NE80.3)
	Cause	Other health care related causes of injury or harm (PL10)
	Mode	Mismatched blood used in transfusion (PL14.3)
	Cluster	NE80.3/PL10/PL14.3
10	Case	Right sided pneumothorax caused by mechanical ventilation in an intensive case setting
	Harm	Pneumothorax, unspecified (CB21.Z); Laterality (use additional code, if desired) - Right (XK9K)
	Cause	Ventilation associated with injury or harm in therapeutic use (PK81.0)
	Mode	Unspecified mode of injury or harm associated with a surgical or other medical procedure (PL11.Z)
	Cluster	CB21.Z&XK9K/PK81.0/PL11.Z
11	Case	Patient with neutropenia due to chronic therapy with clozapine.
	Harm	Acquired neutropenia (4B00.01)
	Cause	Drugs, medicaments or biological substances associated with injury or harm in therapeutic use (PL00); Medication (use additional code, if desired) - XM8UG6]
	Mode	Drug-related injury or harm in the context of correct administration or dosage, as mode of injury or harm (PL13.2)]
	Cluster	4B00.01/PL00&XM8UG6/PL13.2

Note that in each of these examples, a mode/mechanism of harm is coded alongside the cause of harm code for all cases. This is true, even when a mode of harm is not apparent. In the latter situations, a code for 'mode or mechanism of injury unspecified' should be selected, for any of substance-related harm, procedure-related harm, or device-related harm. For the 'other health care related causes' the harm should be coded (from anywhere in the classification) followed by code [PL10](#) *Other health care related causes of injury or harm* followed by the appropriate code from category [PL14](#) *Mode of injury or harm associated with other health care related causes* (where there are several mode options).

Considerations around distinguishing poisoning versus overdose of drugs, medicaments or biologic substances in the clinical context

It is important to make a distinction between an overdose in the context of clinical care and a poisoning that is not in a clinical context. The former would be coded using codes in the 'Causes of health care related harm or injury' section of Chapter 23, whereas poisonings would be coded in the 'Unintentional causes' or 'Intentional self-harm' sections of Chapter 23.

The following scenarios will help to illustrate the distinction:

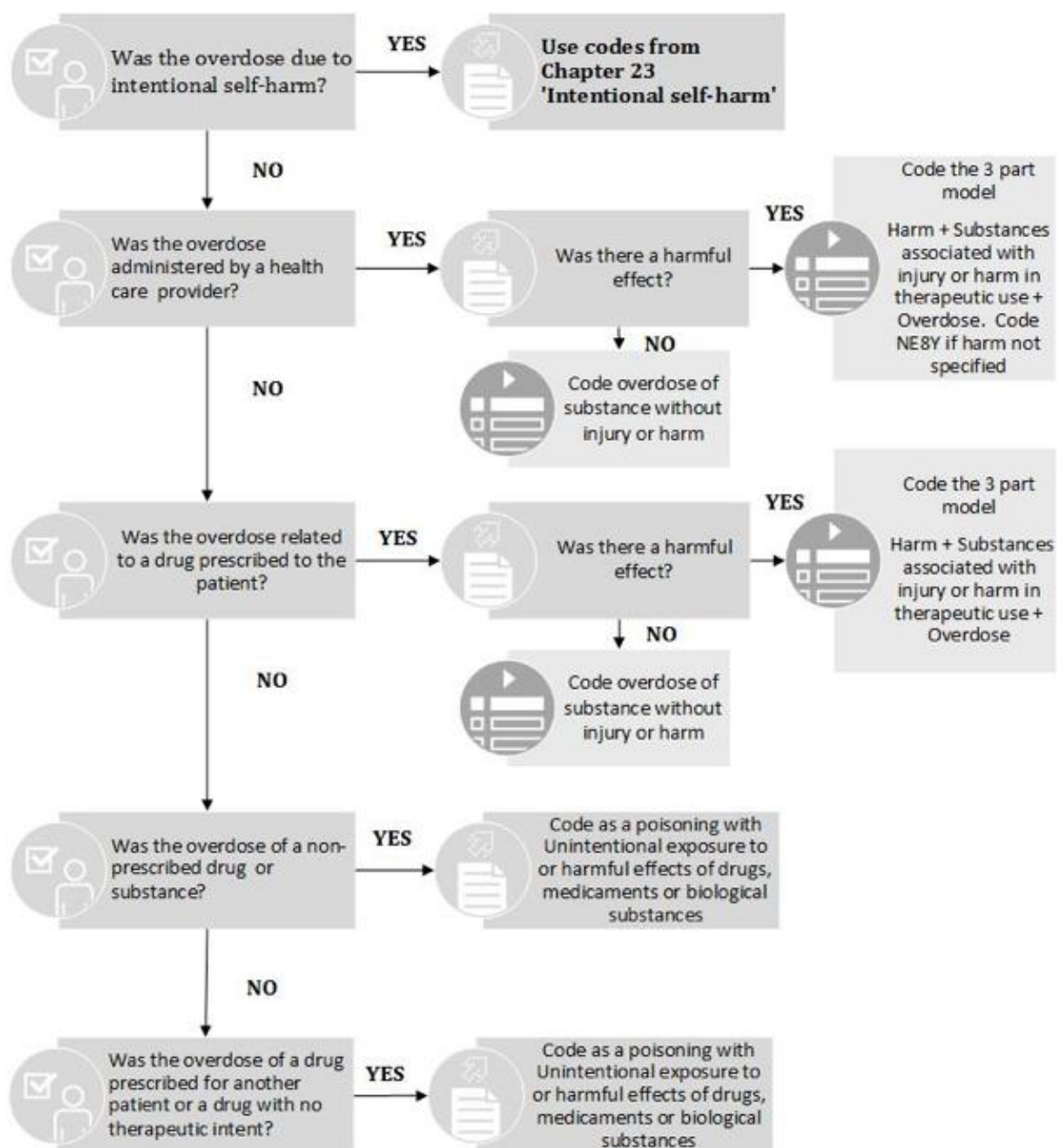
1. An adult medical inpatient receives an overdose of a prescribed medication, because an excess dose is inadvertently injected by a nurse.
2. An adult inadvertently takes an overdose of their own prescribed medication, because the physician wrote the prescription incorrectly.
3. An adult inadvertently takes an overdose of their own prescribed medication, because they were given incorrect instructions by the pharmacist.
4. An adult inadvertently takes an overdose of their own prescribed medication, because they misunderstood instructions on the pill bottle and verbal instructions given to them by both the pharmacist and their doctor.
5. An adult inadvertently takes an overdose of their own prescribed medication, and it is unclear from documentation or case investigation as to why a mistake was made.
6. An adult takes an overdose of their own prescribed medication with undetermined intent.
7. An adult intentionally takes an overdose of their own prescribed medication with intent for self-harm.
8. A child ingests a number of pills from his mother's prescribed pill bottle and becomes somnolent.

Scenario 1 is clearly an overdose arising from an error in a health care context, while scenario 8 is clearly a poisoning of a child who is not in a therapeutic health care context.

Scenario 7 should also be coded as an episode of poisoning, because the pills were not taken with a therapeutic intent, but with intent for self-harm (the 'Intentional self-harm' concept overrides other considerations).

Scenarios 2 and 3 are overdoses arising from problems in a health care context (and are coded using the 'Causes of health care related harm or injury' codes). In both scenarios, the context is one of medication treatment, and the actions of health care providers.

Scenarios 4, 5 & 6 are less straightforward, though rather common in patient care. The context of the medication use is still clearly that of treating a medical condition and the fact that the medication was prescribed to the patients makes it a context of therapeutic use (provided there is no mention of intentional self-harm). Because of the therapeutic context, these scenarios should be coded using the 'Causes of health care related harm or injury' codes, rather than poisoning codes.



Overdose Flowchart

Figure 1: Flowchart for coding poisoning versus overdose.

Instructions on when the three-part quality and safety model applies, and when it does not

The above sections and examples describe scenarios in which an aspect of care (a drug, procedure, device, or other aspect of care) has been causally linked to a condition that a patient has developed. In many instances, however, conditions arise in the health care setting without explicit documentation suggesting a causal link to an aspect of care. Specific examples include:

- pulmonary embolism arising two days after a surgical procedure
- atrial fibrillation after surgery
- low blood pressure one day after administration of a drug
- pneumonia developing on day four of a hospital stay
- urinary tract infection arising in hospital without any mention of catheters

In each of these examples, the three-part model for quality and safety would NOT apply if there is no explicit documentation asserting a causal link to another aspect of care, whether that is a drug, procedure, device, or other aspect of care. Importantly, the mere mention of a surgical procedure or a drug administration in the above examples, does NOT mean that those factors played a causal role, because the clinical statements merely declare timing of the diagnosis, with descriptive words like ‘after’, ‘following’, ‘occurring on day XX’. In such cases, the correct coding of the conditions would be to code the medical condition from any chapter from ICD-11 along with an extension code for timing (in particular, the extension codes for diagnoses arising during a hospital stay, plus or minus the optional extension codes for intraoperative or postoperative timing of a diagnosis).

The above examples would be coded in the following way:

- [BB00.0](#) Acute pulmonary thromboembolism & [XY69](#) Developed after admission & [XY7V](#) Postoperative
- [BC81.3](#) Atrial fibrillation & [XT5R](#) Acute & [XY69](#) Developed after admission & [XY7V](#) Postoperative
- [BA2Z](#) Hypotension, unspecified & [XY69](#) Developed after admission
- [CA40.Z](#) Pneumonia, organism unspecified & [XY69](#) Developed after admission
- [GC08.Z](#) Urinary tract infection, site and agent not specified & [XY69](#) Developed after admission

2.23.20.2 Causation in the context of quality and safety

There are nuances of language in documentation that will indicate whether there is a causal link between a cause and harm.

Connecting terms implying a causal relationship

A causal relationship is strongly suggested by the following terms:

Terms	Additional Notes
as (a) complication of, complicated by, complicating, complication(s) of	-
as a cause of, cause of, caused, caused by, causing	-
as a result of, resulted in, resulting in, with resultant, with resulting	-
because of	-
due to	-
from	-
induced, induced by	-
leading to, led to	-
related to,	-
precipitated by	-
producing	-
secondary to	-
likely related to	Coding judgment call. However, the clinician is making a causal inference with this term
possibly secondary to, probably secondary to	Coding judgment call. However, the clinician is making a causal inference with this term
may be the reason for	Coding judgment call. However, the clinician is making a causal inference with this term

Connecting terms where the causal relationship is unclear

Occasionally there may be connecting terms that hint at causation, but without explicit assertion of a causal link. Examples are shown below. In these circumstances, coders need to check with the documenting clinician, or look for supplementary wording or ancillary information that implies causation.

Terms

Associated with

Accompanied by

Incidental to

Connecting terms NOT implying a causal relationship

In clinical documentation, terms are often used to describe a temporal association. The many terms listed in the preceding table (from 'connecting terms implying a causal relationship') are connecting terms that do suggest a causal association that is typically also a temporal association. In contrast, there are a number of terms that describe only temporal associations. Examples of such terms are listed below:

Terms

after

also

and

during

with

arising in or during

consistent with

followed by, following

incurred after/during/in/when

occurred after/during/in/when/while

postoperatively, postoperative, occurred post-op

If connecting terms of this sort appear in clinical documentation without any of the causal connectors discussed earlier, avoid using the three-part quality and safety model.

Terms like ‘postoperative’, ‘post-op’, ‘postprocedural’, etc., are a special situation because these have historically been considered, in some coding systems to be indicative of a causal link. However, as noted in the specific examples above, conditions such as urinary tract infection, pneumonia, and atrial fibrillation may temporally arise after surgery, without necessarily being caused by surgery. It is for this reason that the guidelines presented here instruct coders to look for explicit causal connections. (Importantly, postoperative conditions such as pneumonia, urinary tract infection, and atrial fibrillation can still be coded with informative extension codes that specify timing – i.e. ‘arising during hospital stay’ and/or ‘postoperative’—and permit the derivation of adverse events in indicators in data analysis).

Other specific situations where the clinical context implies a causal relationship

There are some clinical situations where there may not be connecting terms that explicitly point to causation, but where the clinical circumstances nevertheless clearly point to causation. Some examples appear below:

Specific situations

failed device

infected device

loose screws

postprocedural bleeding

post-op wound infection

dehiscence

wound hematoma

In each of these, it is clear that the situation would not have occurred in the absence of a procedure or a device problem. Accordingly, the three-part quality and safety model should be applied.

In contrast, conditions such as postoperative pneumonia or postoperative pulmonary embolism, or postoperative atrial fibrillation are different than the specific situations listed in the table above. This is because problems such as pneumonia, pulmonary embolism, or atrial fibrillation can be triggered by factors unrelated to the surgical procedure (i.e. different from a 'wound' that is without question caused by surgery).

2.23.20.3 Chronic postprocedural conditions

There are many chronic clinical conditions that occur either as a consequence of specific procedures and techniques or as a result of the removal of an organ, e.g. postmastectomy lymphedema syndrome, postirradiation hypothyroidism. In many instances, codes for such chronic postprocedural conditions are in ICD-11 within various body system chapters.

Examples include:

- [BE10](#) Postcardiotomy syndrome
- [5D40.Z](#) Postprocedural hypothyroidism, unspecified
- [GC72](#) Postprocedural urethral stricture
- [GC70](#) Postoperative adhesions of vagina

These are, by their very nature, precoordinated codes that capture both the a clinical condition and the notion of it being caused by a procedure. It is possible to use such codes on alone without any clustering. However, coders can use the three-part model with such codes to add specificity. The model allows the addition of more specificity about the specific type of surgical procedure that caused the condition, and also the mode through which the procedure caused the condition.

Example 1: Urethral stricture due to previous radiation for treatment of prostate cancer.

Code to [GC72](#) Postprocedural urethral stricture Further detail can be added to the code [GC72](#) with the addition of: [PK81.C](#) *Radiation therapy associated with injury or harm in therapeutic use* and [PL11.Y](#) *Other specified mode of injury or harm associated with a surgical or other medical procedure* Cluster: [GC72/PK81.C/PL11.Y](#)

Note, however, that there will occasionally be instances where it is entirely unnecessary to use the three-part model because the code for the chronic postprocedural condition already contains full clinical detail. For example:

- [9D21](#) *Cataract lens fragments in eye following cataract surgery* (for this, it would be highly redundant to code “ophthalmic procedure” and a corresponding mode, given all the detail that is inherently embedded in this single code).
- [EL61](#) *Chronic radiodermatitis following radiotherapy* (again, redundant to code “radiation therapy” as the procedure causing harm, and “mode unspecified” for this case).

In relation to the two preceding examples, we reiterate that the overriding recommendation is that the three-part model should be used whenever possible. Coders must simply make a case by case judgement when it is obvious to them that the added procedure code and mode code are redundant.

2.23.20.4 Adverse events and circumstances in health care that do not cause actual injury or harm

There are instances in the context of health care where things happen to patients, and where problems arise, but where there is no actual adverse consequence to the patient as a recorded medical condition. Specific examples include:

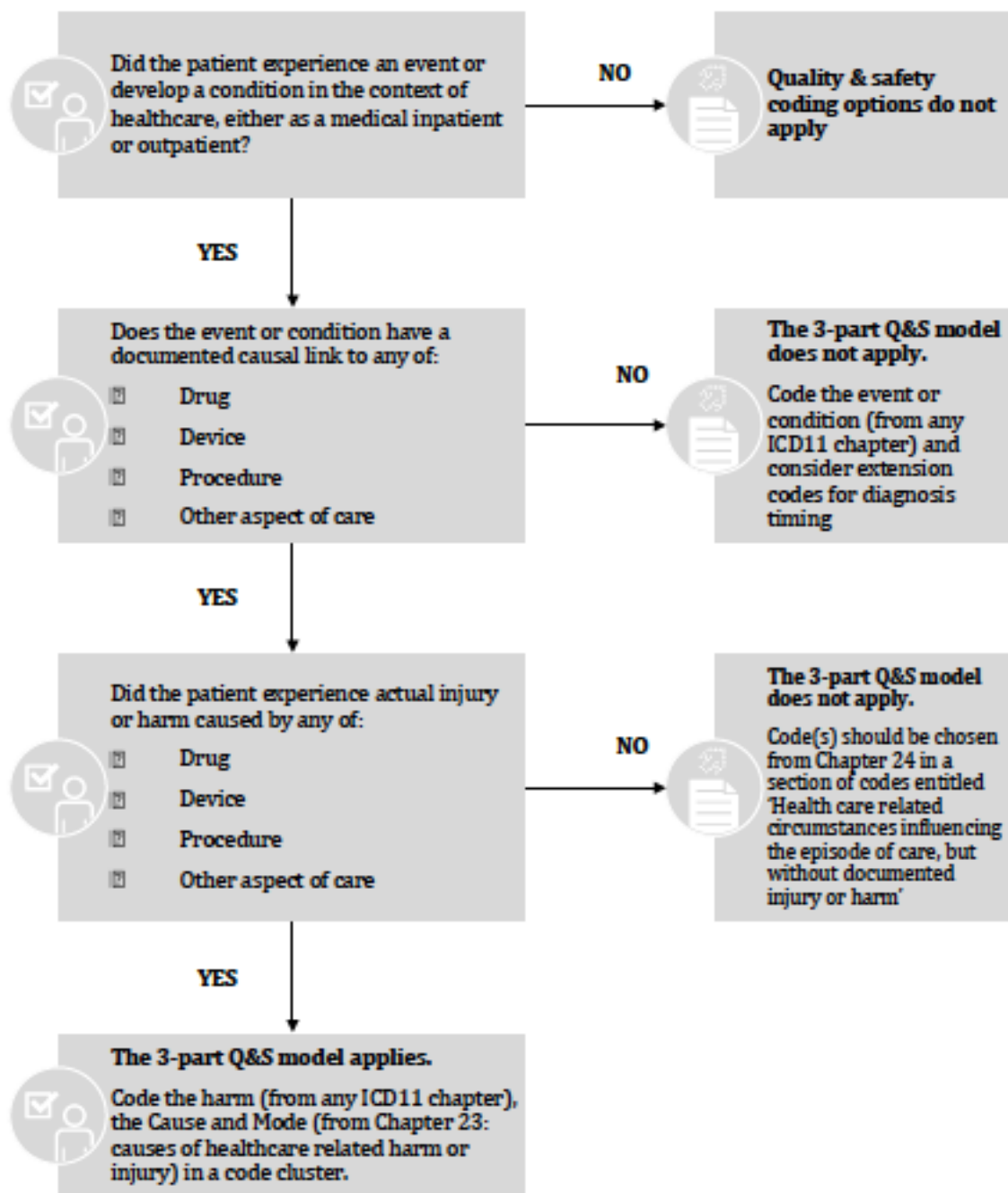
- A fall in the health care setting without fracture or other injury
- An incorrect drug administered without harm to patient
- A drug given to the wrong patient without harm to the patient
- A delay in drug administration without negative effect on clinical course
- Documented failure of sterile precautions in a surgical procedure without ensuing infection
- Dislodged orthopaedic device without symptoms or problems
- Inadvertent needle stick without documented injury or other harm

In these circumstances, codes should be chosen from Chapter 24 Factors influencing health status or contact with health services in the section of codes entitled 'Health care related circumstances influencing the episode of care, without documented injury or harm'. These codes are organised using the four categories of health care related harm that appear in Chapter 23 External causes of morbidity or mortality (drugs, devices, procedures and other health care related causes), but with the important distinction that the circumstances being described through coding did NOT cause actual harm to the patient.

The above examples would be coded in the following way:

- [QA8E](#) Fall in health care without injury or harm
- [QA72](#) Incorrect substance without injury or harm
- [QA8D](#) Patient received diagnostic test or treatment intended for another patient without injury or harm
- [QA8B](#) Delayed treatment without injury or harm
- [QA52](#) Failure of sterile precautions without injury or harm
- [QA62](#) Dislodgement, misconnection or de-attachment without injury or harm
- [QA8F](#) Needle stick without injury or harm

Figure 1: Summary algorithm for coding events and conditions that arise in the context of health care



Q&S Algorithm

2.23.20.5 Recommendations for data capture and organisation

Information systems must be capable of capturing the three components and marking the three codes as belonging to the same cluster (see also instructions for postcoordination and cluster coding).

2.23.21 Chapter-specific notes

Coder guidance is given below for specific chapters where problems may be encountered in selecting preferred 'main condition' codes. The preceding general guidelines and rules apply to all chapters unless a specific chapter note states otherwise.

2.23.21.1 Chapter 1: Infectious and parasitic diseases

Human immunodeficiency virus [HIV] disease

A patient with a compromised immune system due to HIV disease may sometimes require treatment during the same episode of care for more than one disease, for example mycobacterial and cytomegalovirus infections. Only subcategories for HIV disease associated with tuberculosis and malaria are precoordinated in this block for HIV disease. When another specified HIV-caused disease is documented by the health care practitioner, postcoordinate the HIV-caused disease with the appropriate subcategory for HIV disease as recorded by the health care practitioner.

Example 1:

The patient has HIV disease and is admitted for treatment of Kaposi sarcoma of the soft palate.

Main condition: Kaposi sarcoma due to HIV disease

Kaposi sarcoma is documented as an HIV-caused disease. Therefore, the stem code for Kaposi sarcoma is postcoordinated with the applicable stem code for HIV.

Main condition: [2B57.Y](#) Kaposi sarcoma of other specified primary sites & [XA8HL5](#) Soft palate / [1C62.3](#) HIV disease clinical stage 4 without mention of tuberculosis or malaria.

Example 2:

The patient has HIV disease and is admitted for treatment of toxoplasmosis.

Main condition: Toxoplasmosis due to HIV

Main condition: [1F57.Z](#) Toxoplasmosis, unspecified / [1C62.3](#) HIV disease clinical stage 4 without mention of tuberculosis or malaria

Sepsis with or without septic shock

The concept of sepsis has undergone major changes during the last decades and the current description established and widely accepted internationally in 2016 is that sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection.

Sepsis is not considered to be a disease in itself, but a reaction to an infectious disease which may be of bacterial, viral, fungal or protozoal aetiology. Septic shock is defined as a subset of sepsis in which circulatory, cellular and metabolic abnormalities are profound enough to substantially increase mortality.

A cluster involving a case of documented sepsis should include:

- First, a stem code representing the causing infection (specified or unspecified) and as applicable, an optional extension code for the infectious agent if it is known.
- Second, a stem code for sepsis with or without septic shock depending on the documentation

Note: If the causing infection is documented as generalised or a specific infection is not documented, assign a stem code for greatest level of specificity documented in relation to the infection.

Example 1:

Patient admitted for treatment of pneumococcal pneumonia causing sepsis

Main condition: Pneumococcal pneumonia causing sepsis

Code first the causing infection, [CA40.07](#) *Pneumonia due to Streptococcus pneumoniae* and postcoordinate with the stem code for [1G40](#) *Sepsis without septic shock*

Main condition cluster: [CA40.07/1G40](#)

Example 2:

Patient admitted for treatment of severe influenza A H1N1 causing sepsis.

Code first the causing infection, Influenza [1E30](#) *Influenza due to identified seasonal influenza virus* with optional extension code for ([XN297](#) *Influenza A H1N1 virus*) and postcoordinate with the stem code for ([1G40](#) *Sepsis without septic shock*)

Main condition cluster: [1E30](#) & [XN297/1G40](#)

Example 3:

Patient admitted for treatment of sepsis due to E.coli.

Main condition: Sepsis due to E Coli.

Code first the causing infection. In this example, a specific infection is not documented; therefore a code for ([1C41](#) *Bacterial infection of unspecified site*), is coded with optional extension code for ([XN6P4](#) *Escherichia coli*) and postcoordinate with the stem code for ([1G40](#) *Sepsis without septic shock*).

Main condition cluster [1C41](#)&[XN6P4/1G40](#)

Example 4:

Patient presented with septic shock and died shortly after admission.

Main condition: Septic shock, unknown infection

Code first the causing infection. In this example, a specific infection is unknown; therefore, a code for [1H0Z Infection, unspecified] and then postcoordinate with the code for ([1G41](#) *Sepsis with septic shock*)

Main condition cluster: [1H0Z] / [1G41](#)

2.23.21.2 Chapter 2: Neoplasms

When coding neoplasms, refer to the instructions regarding code assignment at the level of the individual categories, and the use of additional morphological or site descriptions from the extension codes. A neoplasm, whether primary or metastatic, that is the focus of care during a relevant episode of health care, should be recorded.

When the 'main condition' recorded by the health care practitioner is a primary neoplasm and the 'other condition' is a secondary neoplasm (metastasis), code each neoplasm

separately. Do not postcoordinate the stem code for primary neoplasm with the stem code for secondary neoplasm.

When the main condition recorded by the health care practitioner is a secondary neoplasm (metastasis) and the primary neoplasm is no longer present (having been removed during a previous episode of care or where the documentation indicates a personal history of that neoplasm), code the secondary neoplasm (metastasis) as the main condition and separately code as an other condition the stem code for 'personal history of'. Do not postcoordinate the stem code for secondary neoplasm with the stem code for 'personal history of'. (See example below). Also, refer to Section [2.23.16 Coding 'History of' and 'Family history of'](#) for further coding direction.

When the main condition recorded by the health care practitioner is 'Follow-up examination' (a circumstance codable to Chapter 24 'Factors influencing health status or contact with health services') and the 'other condition' recorded is a 'personal history of', code the applicable 'follow-up examination' code as the main condition and separately code the stem code for 'personal history of' as the other condition. Do not postcoordinate the 'follow-up examination' stem code with the stem code for 'personal history of'. Refer to Section [2.23.16 Coding 'History of' and 'Family history of'](#) for coding direction.

Example 1:

A patient is admitted for investigation of a lump in the breast. Investigation concludes a malignancy in the left breast. Mastectomy is performed and histopathology shows an invasive ductal carcinoma which has spread to regional lymph nodes (left axilla). Chemotherapy is planned.

Main condition: Invasive ductal carcinoma

Other condition: Metastases to regional lymph nodes

Procedure: Mastectomy

Code the main condition as invasive ductal carcinoma with optional extension code 'left'. ([2C61.0](#)&[XK8G](#)). Cluster code secondary malignancy in axillary lymph nodes with optional extension code 'left' as other condition ([2D60.3](#)&[XK8G](#)).

Example 2:

Patient who has a history of carcinoma of breast resected two years ago is admitted for a bronchoscopy with biopsy. Investigation revealed secondary carcinoma in lung.

Main condition: Secondary carcinoma in lung

Other conditions: Carcinoma of breast resected two years ago

Procedure: Bronchoscopy with biopsy

Code the main condition as [2D70 Malignant neoplasm metastasis in lung] . Code [QC40.3](#) *Personal history of malignant neoplasm of breast* as another condition and postcoordinate the stem code [2C6Z](#) *Malignant neoplasms of breast, unspecified* to specify the personal history is related to malignant primary breast cancer. Refer to Section [2.23.16 Coding 'History of' and 'Family history of'](#). Main condition: [2D70](#) Other condition: Option 1: [QC40.3](#); Option 2: [QC40.3/2C6Z](#)

Example 3:

Patient is admitted for bladder cancer recheck by cystoscopy. The patient has a history of previously excised bladder cancer. No evidence of recurrence seen.

Main condition: Follow-up examination by cystoscopy

Other conditions: History of bladder cancer